

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Concept Mapping

COURSE NAME:

Cloud Computing and
Big Data Analytics

COURSE CODE:

CSA1519

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)

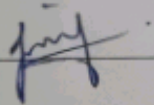
Question Count: 4

Total Marks: 20

Code of Conduct

I Jannathul Firdaya N (192573025) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____



Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

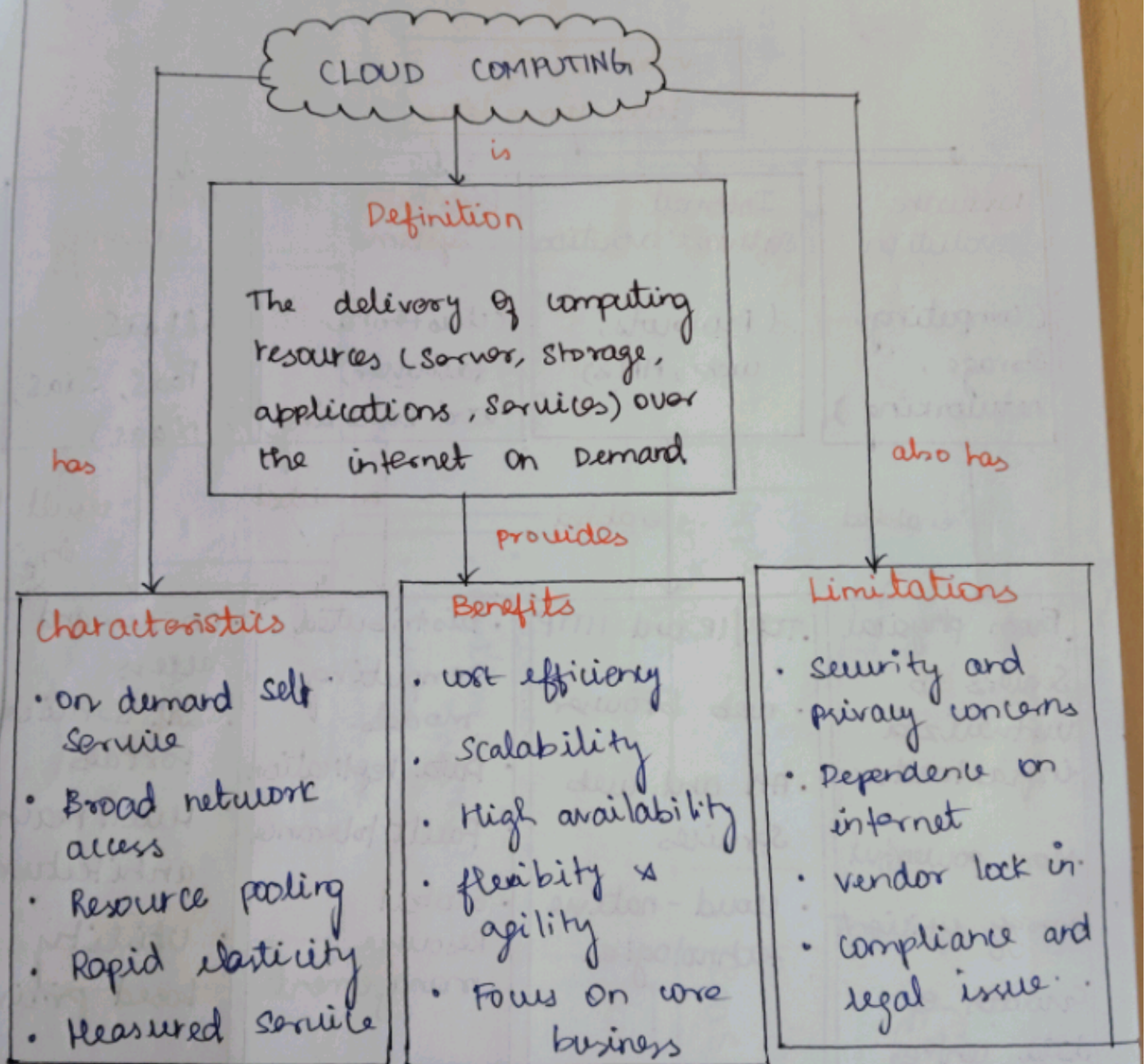
1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

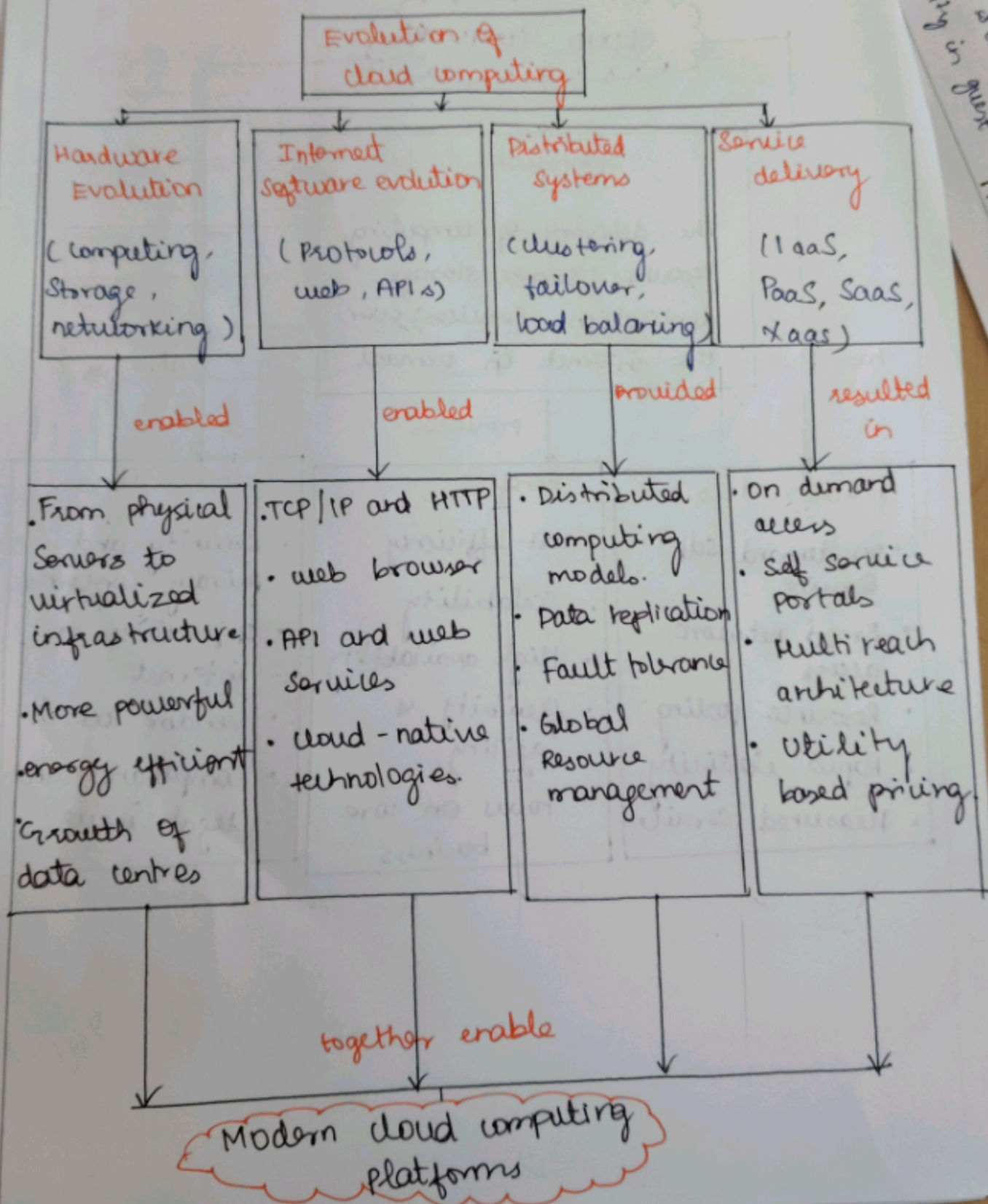
Concept Mapping Rubric					
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

CLOUD COMPUTING FUNDAMENTALS



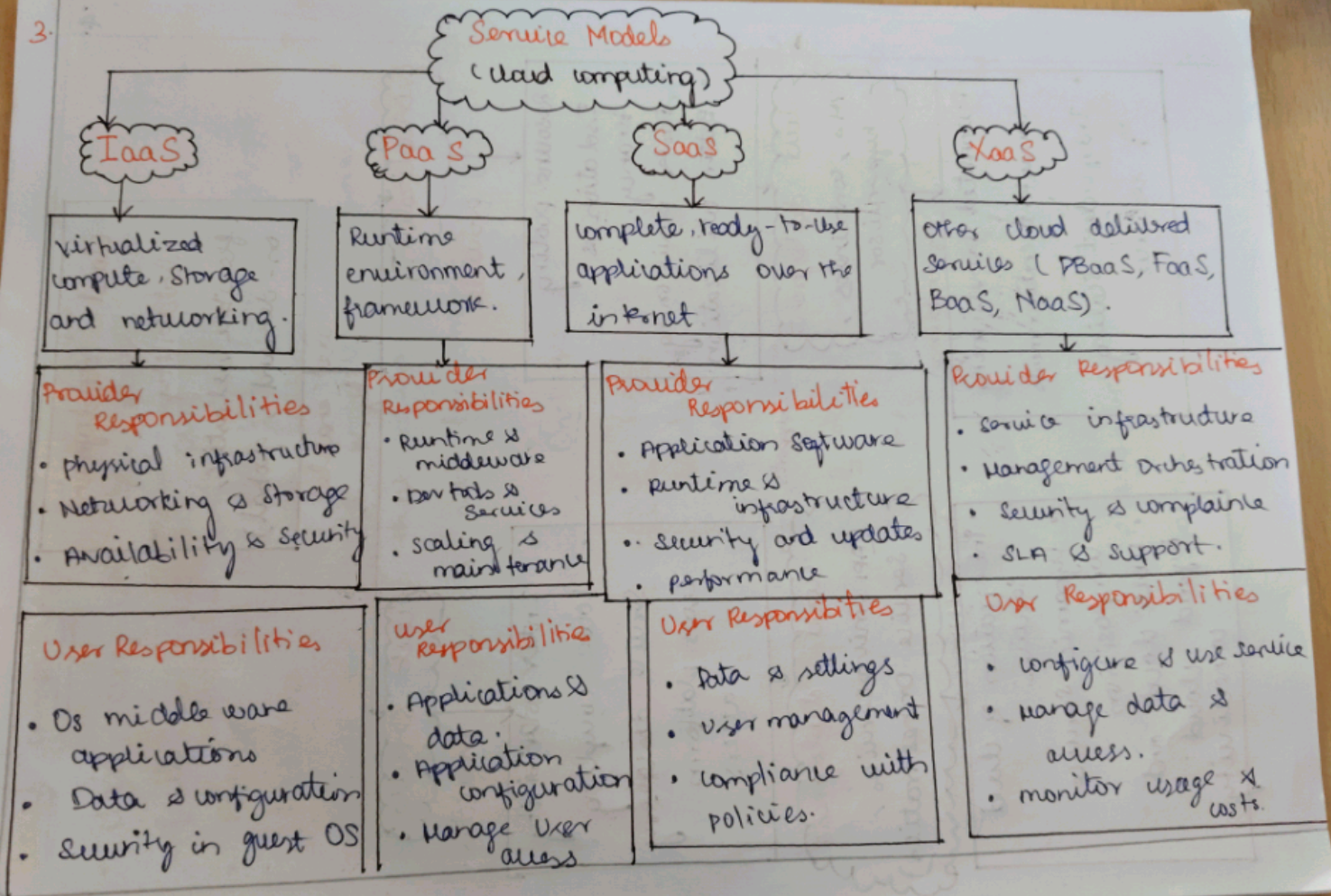
Q.

EVOLUTION OF CLOUD COMPUTING

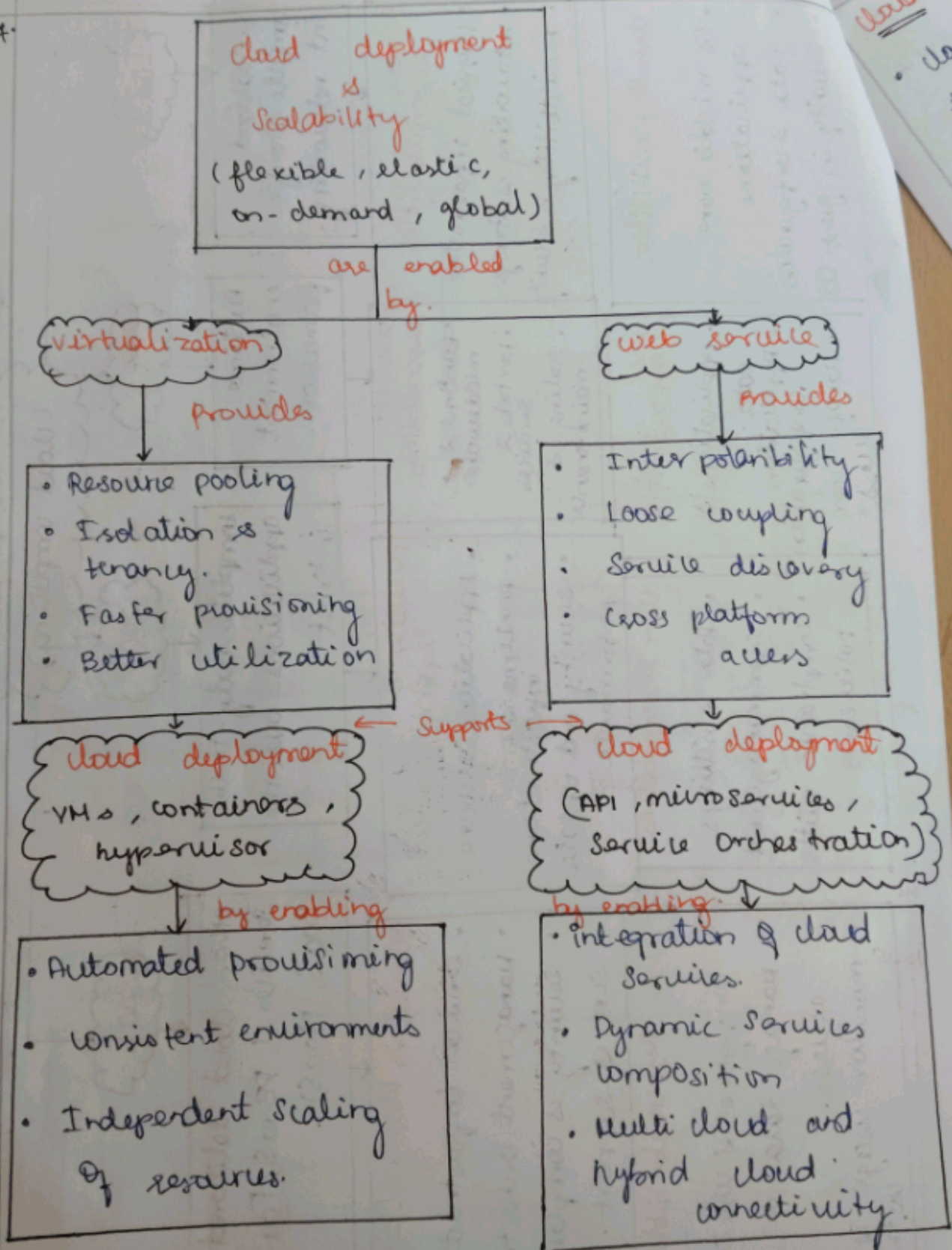


• OS migration
 • Data is configuration
 • Security in guest OS
 • ...

3.



4.



Cloud computing fundamentals.

- cloud computing lets people use computing facilities without owning them.
- Resources can be accessed from different locations using the internet.
- Users can increase or decrease resources according to their needs.
- It reduces the need for maintaining physical computer systems.
- EX: google drive, OneDrive etc.

2. Evolution of cloud computing:

- cloud computing developed gradually from earlier computing technologies.
- powerful computers made it possible to handle larger amounts of data.
- The growth of the internet made remote access easier.
- Distributed computing allowed work to be shared between multiple computers.
- These developments led to today's flexible cloud platforms.

3. Service Models:

- Service models define how cloud resources are provided to users.
- IaaS is useful when user need more control over computing resources.
- PaaS helps developers create and run applications more easily.
- SaaS allow users to directly use software through the internet.
- XaaS provides many different services based on specific user needs.

4. Virtualization and web services as cloud Enablers

- Virtualization allows one physical machine to run multiple virtual systems.
- It helps cloud providers use hardware more efficiently.
- web services allows different applications to communicate with each other.
- together, they make cloud resources easier to manage and expand.
- They help provide flexible services to many users at the same time.